Animesh Joshi

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Education

Purdue University — West Lafayette, Indiana
M.S. in Mathematical Statistics — *August 2026 - May 2028*
B.S. in Mathematics and Statistics — *August 2022 - Dec 2025*

Relevant Coursework: Probability Theory, Statistical Theory, Linear Models, Causal Inference and ML, Multivariate Statistics, Stochastic Processes, Regression Analysis, Machine Learning, Design of Experiments, Operations Research

Technical Skills

Programming: Python, R, SQL, SAS, Bash
Statistics: Regression, Experimental Design, Causal Inference, A/B Testing, Time-Series Analysis, Sequential Analysis
Machine Learning: Supervised Learning, Unsupervised Learning, Neural Networks, LLMs, RAG
Data & Analytics: Pandas, NumPy, SciPy, PySpark, Tableau, Power BI, ETL/Data Pipelines, MS Excel
Tools: Git, AWS, Jupyter, VS Code, RStudio, SSMS

Professional Experience

Software Systems Engineer Intern — May 2025 - Present
Dexcom — *San Diego, CA*

- Designed **sensor simulator** to model glucose-response behavior, eliminating ≈ **$40k+** in hardware costs
- Drove ≈ **30% reduction in producer risk** by identifying and correcting errors in **model selection algorithm**
- Delivered ≈ **20% reduction in consumer risk** through rigorous validation of **statistical tolerance models**
- Designed automated validation system for **signal-processing, nonlinear regression, and statistical inference methods** used for drift characterization, stress-testing **20+** extreme operating conditions
- Built **automated regression testing system** to validate **200+ stored procedures** in manufacturing database

Data Analyst Intern — Jul 2024 - Aug 2024
Axos Bank — *San Diego, CA*

- Engineered **SQL pipelines** to extract loan-level data for 4,000+ recent originations and provide summary tables
- Conducted **cohort analysis for $12.5 billion** in originations over four fiscal years to inform **lending strategy**
- Identified drivers of delinquency by analyzing **longitudinal trends** in credit risk for a **$20 billion portfolio**
- Delivered **Tableau report** highlighting factors contributing to **$500 million in delinquencies** to strategy team

Undergraduate Research Assistant — Aug 2022 - May 2024
Purdue University — *West Lafayette, IN*

- Developed **DL models** for remote water-quality assessment, detecting **70%** of algae hotspots in local lakes
- Designed **automated discrimination detection system** for ML/DL algorithms using **20+** statistical tests
- Implemented **estimation techniques** to quantify the impact of individual data points on overall **model bias**

Selected Projects

Uplift Modeling for Criteo Campaigns
Python, Causal Inference, Machine Learning, A/B Testing

- Conducted **A/B test** to analyze incrementality test and estimate effect of advertisement on conversion rates
- Estimated marginal impact of advertisement on conversion probability using **adjusted Logistic Regression**
- Performed **post hoc power analysis** to ensure adequate sample size for A/B testing and Logistic Regression
- Examined **heterogenous treatment effects** for advertising interventions using T-Learners and Causal Forests

Sequential Experimentation for Email Marketing Campaigns
Python, Sequential Analysis, A/B Testing, Monte Carlo Simulation

- Conducted multi-arm **sequential experiment** analyzing email campaign and compared to fixed-horizon analysis
- Implemented **SPRT** and **alpha-spending methods** to enable early-stopping while avoiding Type I error inflation
- Designed **Monte-Carlo simulation** comparing Type I/II error rate and stopping time across sequential methods
- Estimated **heterogeneous treatment effects** across customer segments using subgroup and interaction analyses

Simulation Study of Tolerance Interval Construction Methods
Python, Monte-Carlo Simulation, Non-Parametric Statistics

- Constructed tolerance intervals using **parametric models, rank-based methods, and density estimation**
- Developed **Monte-Carlo simulation** to estimate coverage percentage and interval width across distributions
- Designed **completely randomized block experiment** to determine best tolerance interval construction method